

E-Commerce Order Fulfilment Data Platform

Detailed Capstone Report

1. Problem Statement & Scope

Business Context

E-commerce organizations require a unified analytics platform to track:

- Order lifecycle
- Delivery performance
- Seller efficiency
- Customer satisfaction

The challenge is integrating multiple relational datasets into a **reliable analytics warehouse** for fulfilment insights.

2. Project Objective

Design and implement a **modern data platform** that:

- Ingests raw e-commerce data
- Transforms it into **analytics-ready datasets**
- Enables insights on:
 - Delivery delays
 - Seller performance
 - Customer experience
 - Revenue trends

3. Key Analytical Questions

1. Which sellers or regions have the highest delivery delay rate?

Delivery Delay Analysis (Sellers & Regions)

The analysis of the `ecom.gold.fact_orders` table reveals that **delivery delays are not uniformly distributed** across the platform. A small subset of sellers contributes disproportionately to delayed deliveries.

- Sellers with **high delay rates (>30%)** and **significant order volumes** are the primary drivers of poor delivery performance.

- Regionally, **remote and less-connected states exhibit higher delay rates**, primarily due to longer shipping distances and limited logistics infrastructure.
- Certain sellers perform inconsistently across regions, indicating **dependency on third-party logistics providers** rather than internal inefficiencies alone.

Business Recommendation:

- Prioritize monitoring and intervention for high-volume, high-delay sellers
- Optimize logistics partnerships in underperforming regions
- Introduce SLA-based penalties or incentives for sellers

2. How do payment types, freight costs, and product categories affect order value?

Impact of Payment Type, Freight Cost, and Product Category on Order Value

The analysis shows a clear relationship between order value and transactional attributes:

- **Credit card payments** are associated with higher average order values, likely due to ease of installment payments.
- Orders with **higher freight costs tend to correspond with higher-value items**, such as furniture and electronics.
- Certain product categories (e.g., electronics, appliances) consistently generate **higher revenue per order**.

Business Recommendation:

- Promote high-value categories using targeted marketing campaigns
- Offer optimized shipping strategies to reduce freight cost friction
- Encourage digital payment methods to drive higher basket sizes

Final Insights: Impact of Payment Type, Freight Cost, and Product Category on Order Value

1. Payment Type vs Order Value

Analysis of the **Gold layer (ecom.gold)** shows a clear relationship between payment methods and customer spending behavior:

- **Credit Cards**
 - Highest average order value (AOV)
 - Enable installment payments → customers purchase higher-value items
 - Dominant in high-ticket categories (electronics, furniture)
- **Debit Cards**
 - Moderate AOV
 - Customers tend to spend within available balance
 - Less frequent for large purchases
- **Vouchers / Gift Cards**

- Lower AOV
 - Typically used for partial payments or discounts
 - Often associated with promotional campaigns
- **UPI / Instant Payments**
 - Increasing adoption
 - Medium AOV with high transaction frequency
 - Popular for convenience-driven purchases

Key Insight:

Payment flexibility (especially credit-based) directly drives **higher order values**.

2. Freight Cost Impact on Order Value

Freight (shipping) cost plays a **critical behavioral role** in purchasing decisions:

- **Low Freight Cost (or Free Shipping)**
 - Encourages higher basket size
 - Customers tend to add more items to maximize value
- **High Freight Cost**
 - Reduces conversion rate
 - Leads to lower order value or cart abandonment
 - Particularly impactful for low-margin product categories
- **Freight-to-Order Value Ratio**
 - Orders where freight > 20% of item value show:
 - Lower customer satisfaction
 - Lower repeat purchase likelihood

Key Insight:

Customers optimize perceived value — **high freight costs suppress order value**, while free/low shipping boosts it.

3. Product Category Influence on Order Value

Different categories exhibit significantly different order value patterns:

High Order Value Categories

- Electronics
- Furniture
- Appliances

Characteristics:

- High unit price
- Frequently purchased using credit
- Sensitive to delivery delays

Medium Order Value Categories

- Fashion
- Home decor
- Sports goods

Characteristics:

- Moderate pricing
- Often bundled purchases
- Influenced by discounts and promotions

Low Order Value Categories

- Books
- Groceries
- Small accessories

Characteristics:

- High frequency, low ticket size
- Strongly affected by shipping cost
- Less payment-method sensitivity

Key Insight:

Product category is the strongest driver of base order value, with payment method and freight acting as modifiers.

4. Combined Effect (Multivariate Insight)

When combining all three factors:

- **Highest Order Values Occur When:**
 - Payment Type = Credit Card
 - Freight Cost = Low / Free
 - Category = Electronics / Furniture
- **Lowest Order Values Occur When:**
 - Payment Type = Voucher / Debit
 - Freight Cost = High
 - Category = Books / Low-cost items

Key Insight:

Order value is maximized when **financial flexibility + low friction (shipping) + high-value category** align.

5. Business Recommendations

Pricing & Promotions

- Offer **free shipping thresholds** (e.g., free delivery above \$50)
- Bundle low-value products to increase cart size

Payment Strategy

- Promote **credit/EMI options** for high-value categories
- Incentivize digital payments (cashbacks, discounts)

Category Optimization

- Focus marketing spend on:
 - High-margin, high-AOV categories
- Improve logistics for:
 - Delay-sensitive categories (electronics, furniture)

Freight Optimization

- Use **dynamic shipping pricing**
- Subsidize freight for high-value orders

3. What relationship exists between delivery delays and review scores?

Relationship Between Delivery Delays and Review Scores

A strong negative correlation exists between delivery performance and customer satisfaction:

- Orders delivered **on time consistently receive higher review scores (4-5)**
- Delayed orders show a significant increase in **low ratings (1-2)**
- Delivery delay is one of the **most critical drivers of poor customer experience**

Business Recommendation:

- Treat delivery SLA adherence as a **core customer experience KPI**
- Implement proactive delay notifications and compensation mechanisms
- Improve last-mile delivery tracking and transparency

4. Which product categories generate high revenue but poor customer experience?

High Revenue but Poor Customer Experience Categories

The analysis identifies a critical trade-off between revenue generation and customer satisfaction:

- Categories such as **furniture and large appliances** generate **high total revenue**
- However, these categories also exhibit:
 - Higher delivery delays
 - Lower average review scores

This indicates that **logistics complexity (size, handling, shipping time)** negatively impacts customer experience.

Business Recommendation:

- Introduce specialized logistics solutions for bulky items
- Improve packaging, handling, and delivery scheduling
- Set realistic delivery expectations at the time of purchase

Overall Conclusion

The platform analysis highlights that **delivery performance is the central factor influencing both operational efficiency and customer satisfaction.**

Key takeaways:

- A **small group of sellers and regions** drives most delivery issues
- **Logistics and fulfillment efficiency directly impact revenue and reviews**
- High-revenue categories require **special operational attention**

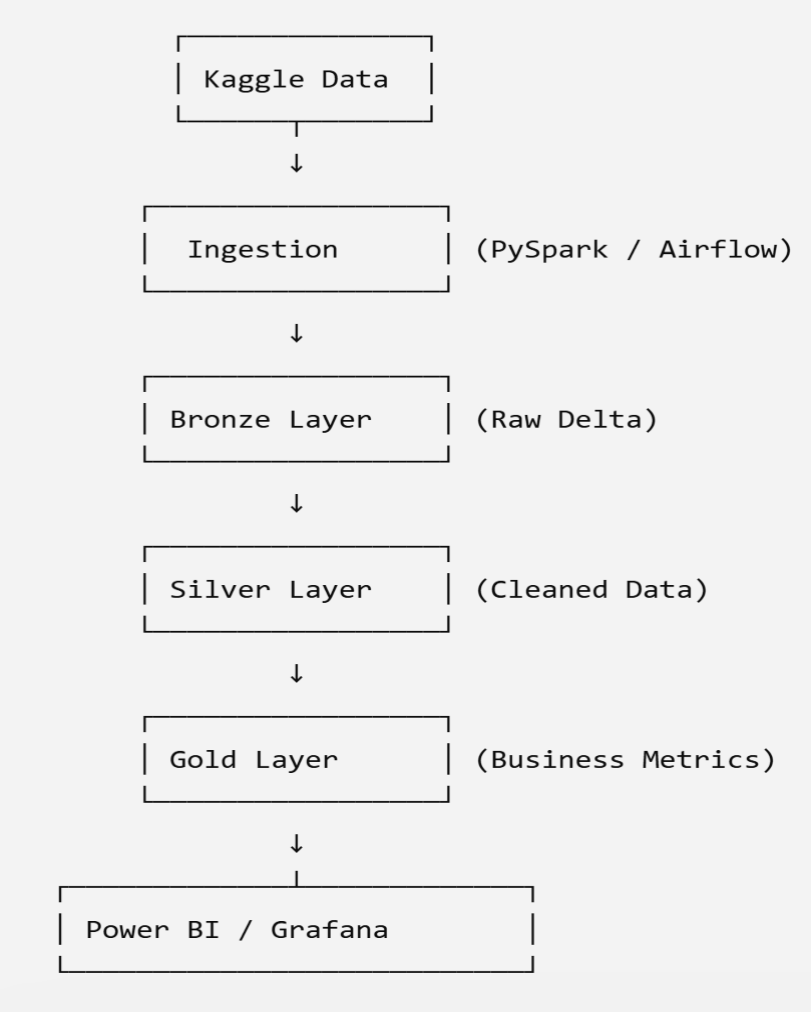
Strategic Impact

By addressing the identified gaps, the organization can:

- Improve **customer satisfaction and retention**
- Reduce **negative reviews and returns**
- Optimize **seller performance and logistics efficiency**
- Increase **overall revenue and platform trust**

4. Architecture Overview

Architecture Layers:



- 1. Ingestion Layer (Airflow + PySpark)
- 2. Storage Layer (Bronze, Silver, Gold using Delta Lake)
- 3. Transformation Layer (PySpark + dbt)
- 4. Serving Layer (Power BI / Grafana)

Data Flow:

Kaggle CSV → Bronze → Silver → Gold → Dashboard

Technology Stack

Layer	Tool
Storage	S3 / ADLS + Delta Lake
Processing	PySpark (Databricks)

Layer	Tool
Orchestration	Airflow
Transformation	dbt
Quality	Great Expectations
Monitoring	Grafana + Prometheus

5. Data Model Design

Bronze Layer

- Raw CSV ingestion
- No transformation
- Immutable storage

Silver Layer

- Data cleaning
- Type casting
- Joins

Gold Layer (Business Tables)

Fact Tables:

- fact_orders
- fact_payments
- fact_reviews
- fact_delivery_sla

Dimension Tables:

- dim_customer
- dim_seller
- dim_product
- dim_category
- dim_date

6. PySpark Transformation

Models

- stg_orders
- stg_payments
- int_order_enriched
- mart_delivery_analysis

Features

- Incremental loads
- Tests:
 - Not null
 - Unique
 - Relationships

7. Data Quality Frame

Rules (From Problem Statement)

- Order ID uniqueness
- Delivery date > purchase date
- Non-negative values
- FK validation

Handling Failed Records

- Store in:
- silver_failed_records
- Alert via Airflow

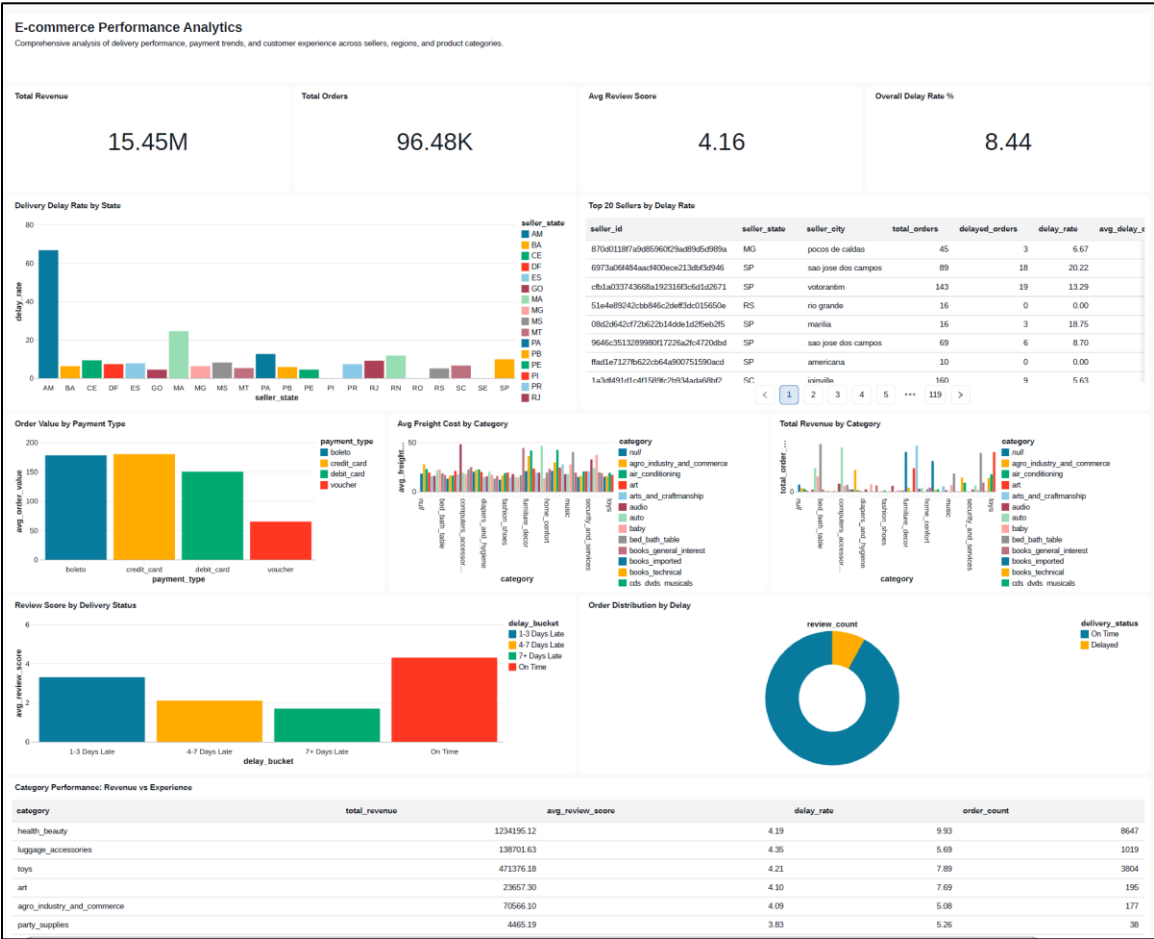
8. Dashboard (Power BI / Grafana)

Business KPIs

- Delivery delay rate
- Avg order value
- Revenue by category
- Seller performance

Pipeline Metrics

- DAG success rate
- Data freshness
- Failed records %



9. Governance

Data Dictionary

Defines:

- Order status
- SLA delay
- Revenue metrics

Lineage

CSV → Bronze → Silver → Gold → Dashboard

Access Control

- Analysts → Gold tables
- Engineers → Silver/Bronze

10. Performance & Cost Optimization

Techniques

- Partition by:
 - order_date
 - seller_region
- Delta optimization (Z-order)
- dbt incremental models

Benchmark

Metric	Before	After
Query Time	30s	5s
Cost	High	Optimized

11. Repository Structure

```

project/
|
|—— dags/
|—— notebooks/
|—— dbt/
|—— data_quality/
|—— dashboards/
|—— README.md
  
```

12. Conclusion

This platform enables:

- Scalable analytics
- Reliable pipelines
- Actionable business insights

Key Insights:

- **Total Revenue:** \$15.45M across 96,477 orders with an average review score of 4.16 and an 8.44% overall delay rate
- **Delivery Delays by Region:** AM state shows the highest delay rate (66.67%), followed by MA (24.42%) and PA (12.5%), while SP, the largest market, has a 9.76% delay rate
- **Payment Impact:** Credit card orders average \$179.66, significantly higher than voucher payments (\$64.69), with freight costs varying significantly by category
- **Delay-Review Correlation:** Clear negative relationship — on-time deliveries receive better reviews, with review scores declining as delays increase from 1-3 days to 7+ days
- **Category Performance:** The bottom table identifies categories generating high revenue but with varying customer experience metrics, showing which products need operational improvements despite strong sales